

Exploratory Analysis: Discography and Tracks

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1 Introduction

This report describes the exploratory analysis of the canonical discography of the Neomelodic Corpus (32,508 title-level tracks for 190 artists¹). The analysis focuses on productivity, release formats, the evolution of track duration, and a first exploratory semantic analysis based on titles.

2 Data Quality and Artist Integration

Track data were integrated with artists' biographical metadata after a two-stage cleaning process (geographic inference and homonym resolution). This ensures that productivity analyses by region and generation are not distorted by matching errors (e.g., international homonyms).

2.1 Missing-Data Analysis

The discographic dataset presents the following critical issues:

- **Duration:** in the raw CSV, duration is missing for 8,621 records; in the deduplicated and cleaned dataset used in the analyses, duration remains missing for 6,392 tracks. Most missing values come from Discogs sources without a specified running time.
- **Release year:** high coverage (>95%) thanks to the integration of multiple sources.

3 Discographic Data Provenance

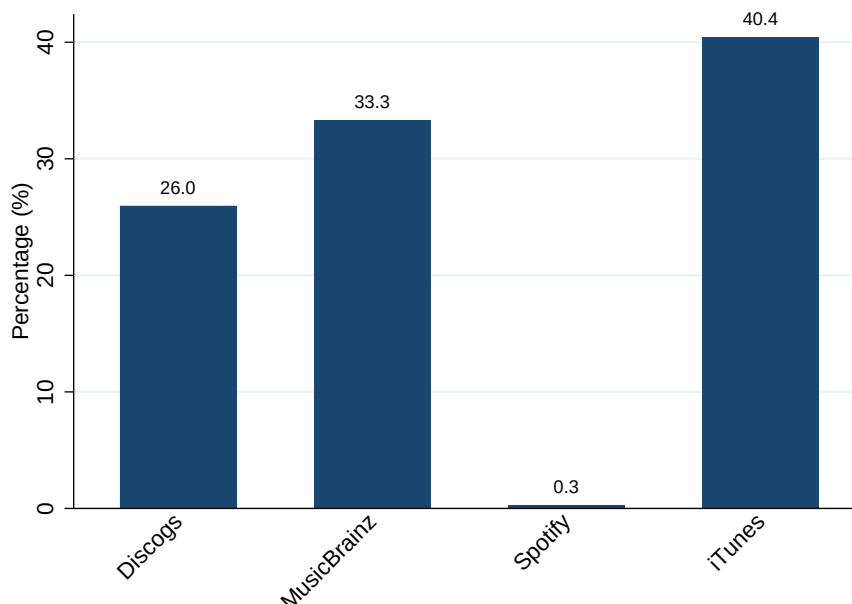
Rationale: We verify the distribution of source platforms to assess source heterogeneity and the robustness of the consolidated database.

Measures: Percentage distribution of tracks by primary source.

¹The raw relational database includes a total of 45,815 discographic records, corresponding to individual tracks by release or album. For the analysis presented here, however, we use the canonical layer `tracks_unique_title_level.dta`, which collapses release-level duplicates and the main title variants, reducing the population to 32,508 title-level tracks. This choice is more coherent than the old `duplicates drop name track_title, force`, but it should still not be read as a perfect ontological identification of “unique works,” because it does not exhaustively distinguish live versions, medleys, or substantive reuses of the same song.

Interpretation: Figure 1 shows that the consolidated dataset draws its structure mainly from **iTunes (38.8%)**, **MusicBrainz (32.8%)**, and **Discogs (28.1%)**. The share classified in the “Spotify” bucket in the live dataset used by the pipeline is residual (**0.3%**). The correct reading of this figure is therefore multi-platform metadata provenance, not substantive dependence on Spotify.

Figure 1: Distribution of Discographic Data Sources (Percent)



Notes: The Y-axis reports the percentage over the total of 32,508 canonical title-level tracks. In the live dataset used by the pipeline, iTunes, MusicBrainz, and Discogs account for almost the entire provenance of the metadata; the “Spotify” bucket is residual.

Stata code: 01.exploratory_tracks.do, lines 246–266. `replace source_clean = ...; collapse (sum) count=one, by(source_clean); gen pct = (count/total)*100; graph bar pct ...`

4 Productivity Analysis

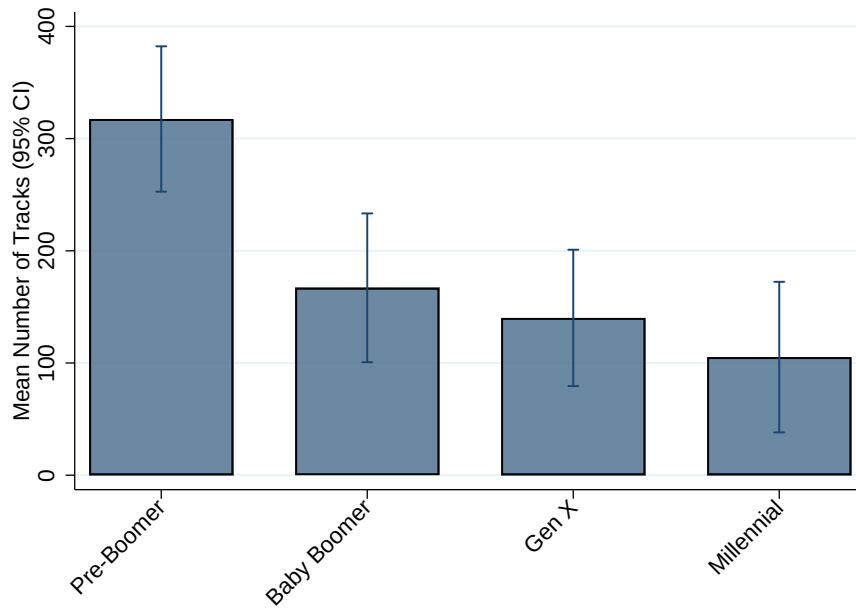
4.1 Productivity by Generation

Rationale: Measuring track output capacity for each main generational cohort makes it possible to assess artistic longevity and career intensity over time.

Measures: Mean number of tracks per artist grouped by generational cohort.

Interpretation: Figure 2 shows a clear productivity hierarchy linked to career seniority. *Pre-Boomers* display the highest average volume, followed by *Baby Boomers* and *Gen X*, while *Millennials* are on average less prolific in the current dataset. The most recent cohorts should still be read cautiously because of their smaller size.

Figure 2: Average Number of Tracks per Artist (Four Main Cohorts)



Notes: The Y-axis indicates the mean number of tracks per artist with 95% confidence intervals. The panel excludes Gen Z and Unknown; it concerns only Pre-Boomer, Baby Boomer, Gen X, and Millennial.

Stata code: 01_exploratory_tracks.do, lines 117–136. `keep if inlist(gen_order,1,2,3,4); collapse (sum) total_tracks=one, by(name gen_order); marginsplot ...`

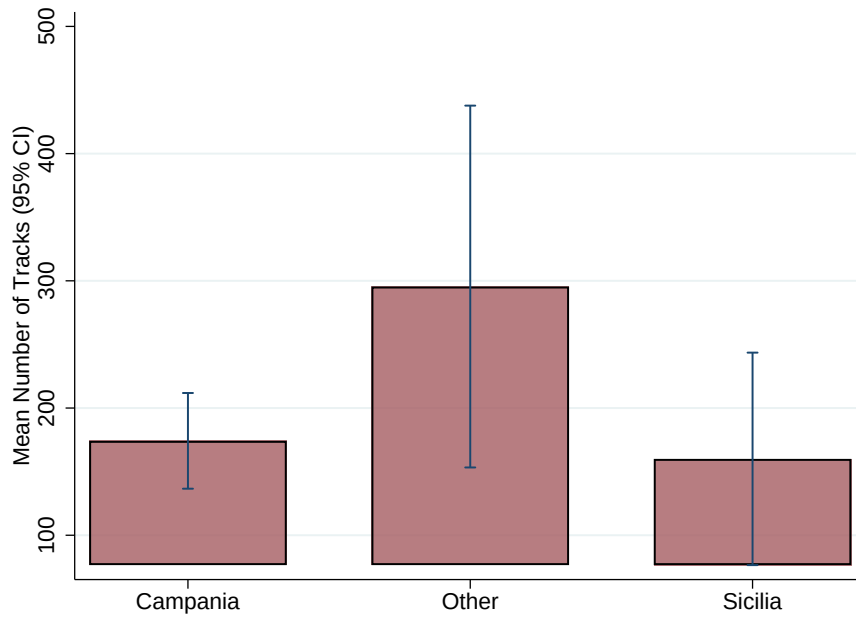
4.2 Productivity by Macro-Region

Rationale: Given the genre’s concentration in Campania and Sicily, we aggregate other regions into an “Other” category in order to compare the “core” against the neomelodic diaspora.

Measures: Total track volume and mean per artist by regional group.

Interpretation: Although Campania dominates in absolute volume, the average per artist in the “Other” category (Figure 3) is higher (about 314 tracks per artist). This phenomenon should be read cautiously, because the “Other” group contains only 10 artists and is therefore highly sensitive to a few very prolific cases.

Figure 3: Average Number of Tracks per Artist (Core vs Other)



Notes: Comparison between historical poles (Campania, Sicily) and diaspora (Other). The high mean in “Other” is driven by a few isolated but prolific artists, as suggested by the wide confidence intervals.

Stata code: 01_exploratory_tracks.do, lines 138–155. `collapse (sum) total_tracks=one, by(name region_group); margins region_id; marginsplot ...`

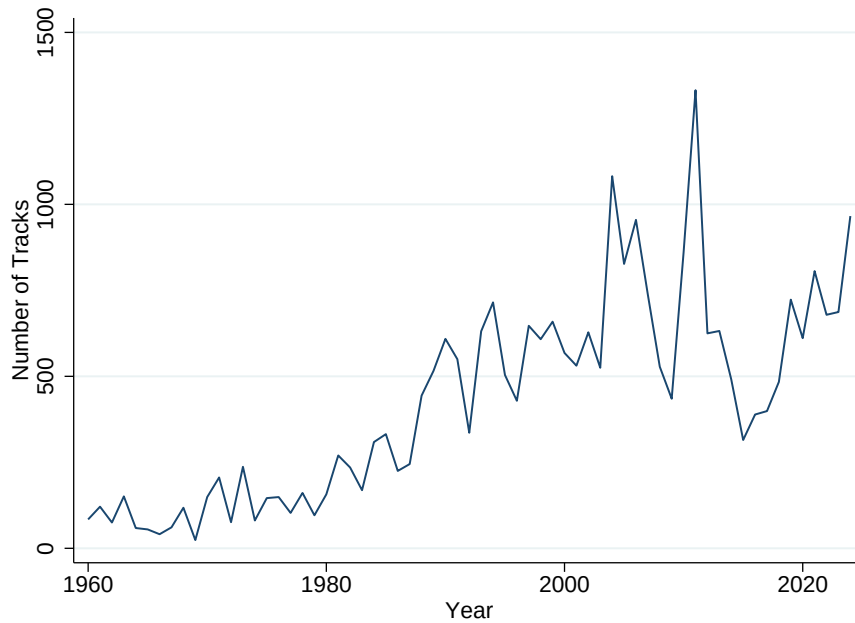
5 Temporal and Technical Trends

Rationale: The analysis of temporal trends makes it possible to map the evolution of the neomelodic market in response to technological innovations (from vinyl to digital).

Measures: Annual track volume and release-format frequency.

Interpretation: As shown by Figure 4, track production remains limited until the late 1990s and then grows sharply in later decades. The chart is descriptive: it is consistent with the transformation of production and distribution channels, but on its own it does not identify the causal mechanisms behind this growth.

Figure 4: Total Number of Tracks Released per Year



Notes: The Y-axis reports the annual volume of new tracks. The trend shows the production boom associated with post-2015 market digitalization.

Stata code: 01_exploratory_tracks.do, lines 201–213. `collapse (sum) track_count=one, by(release_year); line track_count release_year ...`

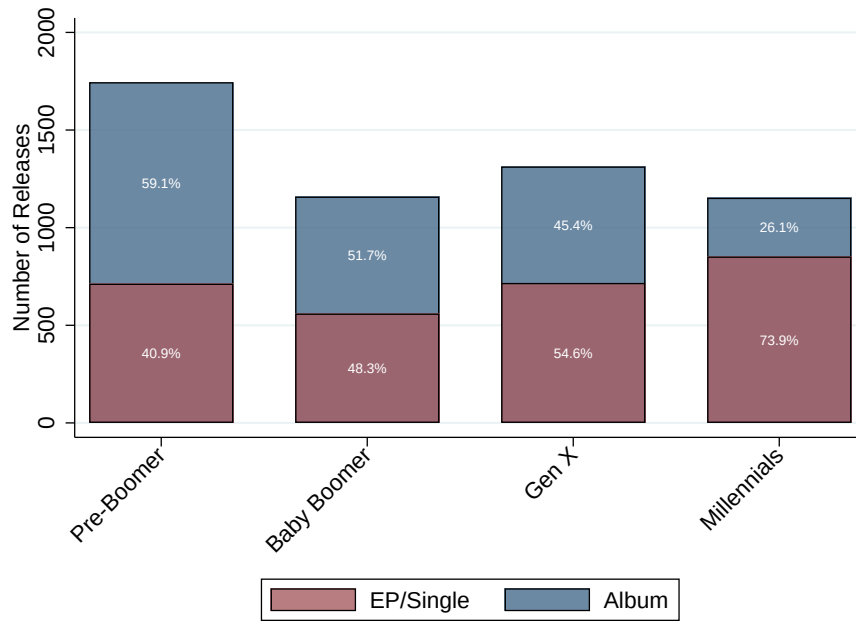
5.1 Distribution Strategies and Formats

Rationale: Identifying the preference for Albums versus Singles/EPs by generation reveals the degree to which artists adapt to new channels of consumption (streaming).

Measures: Binary classification of releases into “EP/Single” (≤ 3 tracks) and “Album” (> 3 tracks), crossed with generational cohort.

Interpretation: The cross-tabulation of formats and cohorts (Figure 5) suggests a greater relative incidence of EPs/singles in more recent cohorts. Again, caution is necessary: the figure should be read as a descriptive pattern rather than robust evidence of a generational break.

Figure 5: Frequency of Albums vs EPs/Singles by Generation (Four Main Cohorts)



Notes: Distribution of release formats. The Y-axis reports the absolute number of releases, while the internal white labels indicate the percentage of the format relative to the generation total.

Stata code: 01_exploratory_tracks.do, lines 157–199. `duplicates drop name release_title gen_order rel_type, force; keep if inlist(gen_order,1,2,3,4); reshape wide count ...`

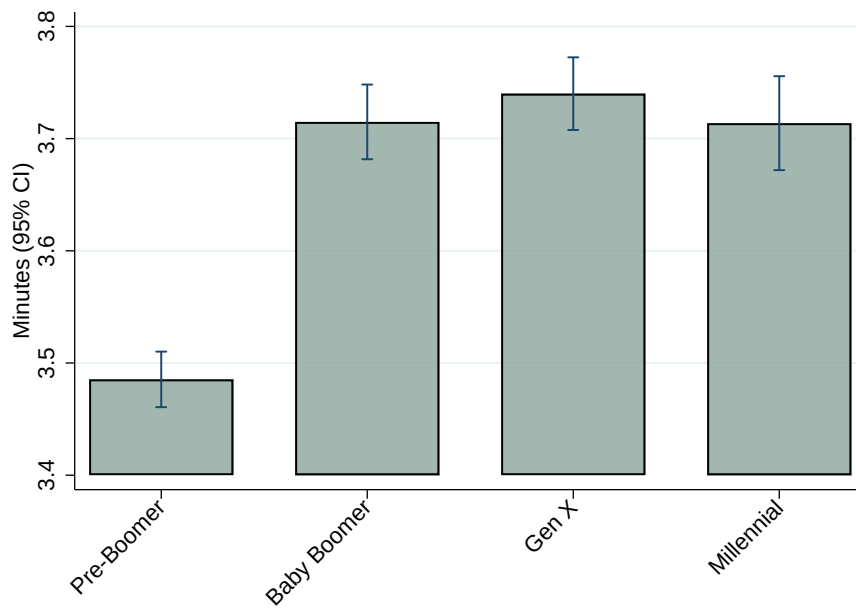
6 Duration Analysis

Rationale: Track duration is a fundamental technical indicator for understanding changes in listening standards and the persistence of extended narrative structures.

Measures: Mean duration (minutes) by generation and historical trend.

Interpretation: The analysis of track duration (Figure 6) includes all release types. No sharp contraction in mean duration emerges across the main cohorts. Differences between extreme cohorts should nevertheless not be overloaded with substantive meaning without a broader base.

Figure 6: Average Track Duration (Minutes) by Generation (Four Main Cohorts)

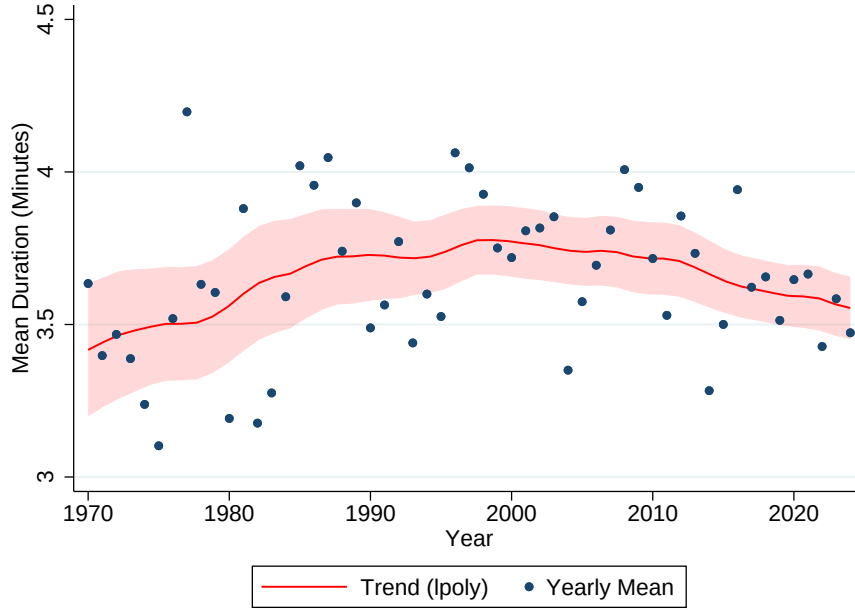


Notes: The Y-axis reports average duration in minutes. The analysis is conducted on all formats, but only for Pre-Boomer, Baby Boomer, Gen X, and Millennial. Bars indicate the mean and the segments the 95% confidence interval.

Stata code: 01_exploratory_tracks.do, lines 269–286. Snippet: `drop if duration_min <= 0.5; keep if inlist(gen_order,1,2,3,4); marginsplot ...`

Trend Interpretation: The historical evolution (Figure 7) shows annual mean values (points) and the trend computed through local polynomial regression (red line) with its 95% confidence interval.

Figure 7: Historical Evolution of Average Duration (1970–2024)



Notes: Blue points represent average annual duration. The red line represents the local polynomial trend (lpoly) with a 95% confidence area.

Stata code: 01_exploratory_tracks.do, lines 288–306. Snippet: collapse (mean) avg_dur=duration_min, by(release_year); twoway (lpolyci ...) (lpoly ...) (scatter ...) ...

7 Semantic Analysis of Titles

The content analysis of the neomelodic corpus is based on the extraction and classification of terms appearing in song titles. This section should be read as an exploratory semantic analysis of titles, not as a definitive classification of song content. To reduce linguistic and dialectal noise, the analysis is preceded by a three-stage text-cleaning protocol:

1. **Text normalization:** conversion to lowercase, systematic removal of complex punctuation (including quotes, graphic accents, and ellipses), and elimination of widespread technical and editorial markers in metadata (e.g., *feat*, *live*, *remix*, *prod*, *version*, *bonus*, *edit*).
2. **Ultra-aggressive lexical filtering:** removal of grammatical words with no informative value (articles, pronouns, prepositions, conjunctions), common temporal adverbs in both Italian and Neapolitan (e.g., *prima*, *dopo*, *mentre*, *quando*), generic and copulative verbs in inflected form (e.g., *essere*, *avere*, *volere*, *stare*), and a broad list of recurrent person names or figures external to genre poetics (e.g., linked to homonyms in classical opera or generic songs).
3. **Phonetic and dialect normalization:** bringing dialectal orthographic variants back to the same semantic root (e.g., the systematic unification of the Neapolitan term *ammore* to the standard lemma *amore*).

The 8 Macro-Categories: The lemmas resulting from this cleaning process were then mapped onto a set of sub-topics aggregated into 8 macro-categories. Because classification is title-based and regex-based, some categories remain more fragile than others. In this revision, the dictionaries for *Malavita*, *Territory*, *Economy*, and *Destiny* were made more conservative, reducing roots that were too generic. Table 6 in the Appendix reports the reference lexical mapping.

- **Love:** includes traditional love songs, romantic passion, and sentimental torment.
- **Malavita:** in the current version it is rendered more conservatively and mainly groups omertà, respect/honor, fugitivity, and explicit references to law enforcement.
- **Family:** collects references to biological and blood ties (mother, father, children, siblings).
- **Economy (Money and Success):** represents the narrative of economic ascent and personal success, with restricted dictionaries to reduce false positives from overly generic words.
- **Death and Mourning:** concerns loss, the tragedy of loved ones’ disappearance, and mourning rituals.
- **Territory:** expresses local belonging through explicit toponyms and territorial references, avoiding overly generic terms such as *sea* or *sun*.
- **Faith:** includes religious references, prayers, saints, and divine figures.
- **Destiny (Life and Fortune):** addresses existential path, fate, and fatality, but in a more selective form than in earlier versions.

Important: The analyzed themes are **not mutually exclusive**. Since classification occurs at the level of the single title, a track titled “Mamma e Carcere” or “Un Amore Lontano” will contain lemmas associated simultaneously with multiple macro-themes (e.g., Family and Malavita, or Love and Malavita), reflecting the intrinsic hybridization of neomelodic poetics.

7.1 Word Clouds

Rationale: To identify the *distinctive* vocabulary of each macro-theme in titles, highlighting the words that characterize it relative to tracks that *do not* belong to that theme.

Method: For each theme T , the size of each word w is proportional to a *distinctiveness score* computed as:

$$\text{score}(w, T) = \left[\frac{f_T(w)}{\sum_v f_T(v)} \bigg/ \frac{f_{\bar{T}}(w) + 0.5}{\sum_v f_{\bar{T}}(v)} \right]^{0.45}$$

where $f_T(w)$ is the frequency of w in titles of theme T and $f_{\bar{T}}(w)$ is the frequency in the complement (tracks *not* classified in T), approximated by the difference between general frequency and thematic frequency. The 0.5 term is Laplace smoothing for rare words; the 0.45 exponent compresses the dynamic range for a visually balanced cloud. Italian and Neapolitan stopwords, proper names, auxiliary verbs, technical musical terminology (*feat*,

remix, live, edit), and English service words are removed. For the **General Corpus**, raw absolute frequency is used.

Interpretation: Figure 8 offers descriptive support for the most distinctive vocabulary of titles. In the general corpus, terms of neomelodic romanticism dominate (*amore, bella, notte, vita*). In some themes, plausible and readable clusters emerge, but this evidence alone is not enough to strongly “validate” the mapping: word clouds help inspect lexical fields, they do not replace a harder semantic validation of the dictionaries.

Figure 8: Word Clouds by Macro-Theme (Titles)



Stata/Python code: 02_content_analysis_tracks.do, lines 177–233, exports cleaned tokens to `top_words_general.csv` and `top_words_<theme>.csv`; `generate_wordcloud.py` rereads the CSVs and applies the distinctiveness score relative to the complement. Snippet: `clean_and.export title.clean ".../top_words_general.csv"`

7.2 Theme Distribution by Gender

Rationale: As in the lyrics report, it is useful to test whether the title-based thematic architecture changes systematically between women and men performers.

Measures: The following table reports, for each macro-theme, the mean share of classified titles by gender, with Standard Error, Women-Men difference, and t-test p-value.

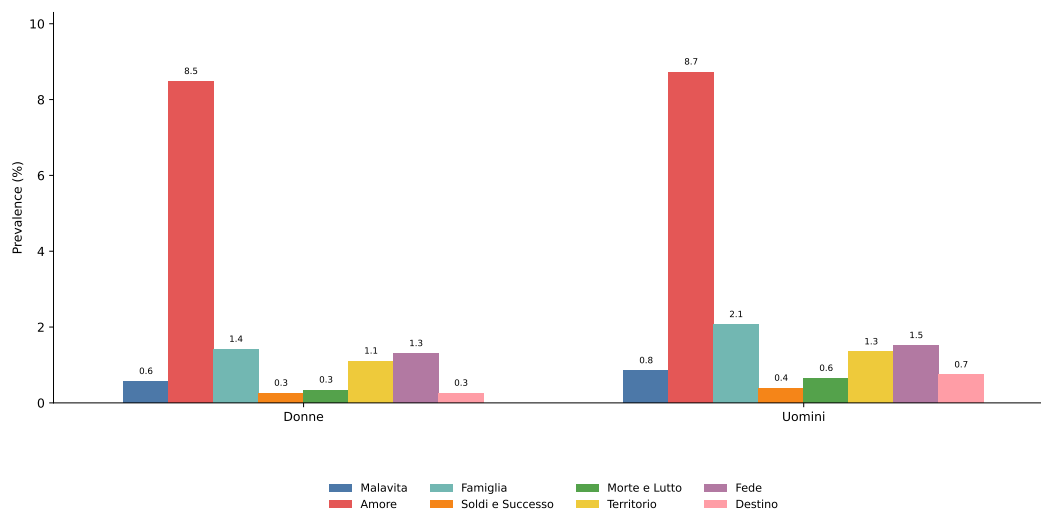
Interpretation: This section mainly serves to separate visual differences from statistically readable differences. In the title-based domain, caution is warranted: themes are binary, rates are low in many categories, and dictionary precision is not uniform.

Table 1: Mean differences by gender for title-based macro-themes

Theme	Women (SE)	Men (SE)	Diff.	p-value
Malavita	0.6 (0.001)	0.8 (0.001)	-0.3	0.038
Love	8.5 (0.005)	8.7 (0.002)	-0.2	0.615
Family	1.4 (0.002)	2.1 (0.001)	-0.7	0.002
Money and Success	0.3 (0.001)	0.4 (0.000)	-0.1	0.151
Death and Mourning	0.3 (0.001)	0.6 (0.000)	-0.3	0.005
Territory	1.1 (0.002)	1.3 (0.001)	-0.2	0.194
Faith	1.3 (0.002)	1.5 (0.001)	-0.2	0.297
Destiny	0.3 (0.001)	0.7 (0.001)	-0.5	0.000

Stata/Python code: `02_content_analysis_tracks.do`, final saving block on the canonical layer `tracks_unique_title_level.dta`; `export_track_report_tables.py` computes gender t-tests and writes `track_theme_gender_table.tex`.

Figure 9: Prevalence of Macro-Themes by Gender



Notes: The chart reports all 8 macro-themes, with separate bars for women and men. The previous table reports pointwise tests. **Stata code:** `02_content_analysis_tracks.do`, lines 141–151. Snippet: `graph bar mt100..., over(sex) asyvars ...`

7.3 Theme Distribution by Generation

Rationale: The analysis of thematic distribution stratified by the main generational cohorts is useful for tracing continuity and discontinuity in title-based vocabulary over time, without overloading cohorts that are too small.

Measures: Percentage incidence of the 8 macro-themes on titles, calculated separately for Pre-Boomer, Baby Boomer, Gen X, and Millennial. The following table also reports pairwise comparison p-values across cohorts.

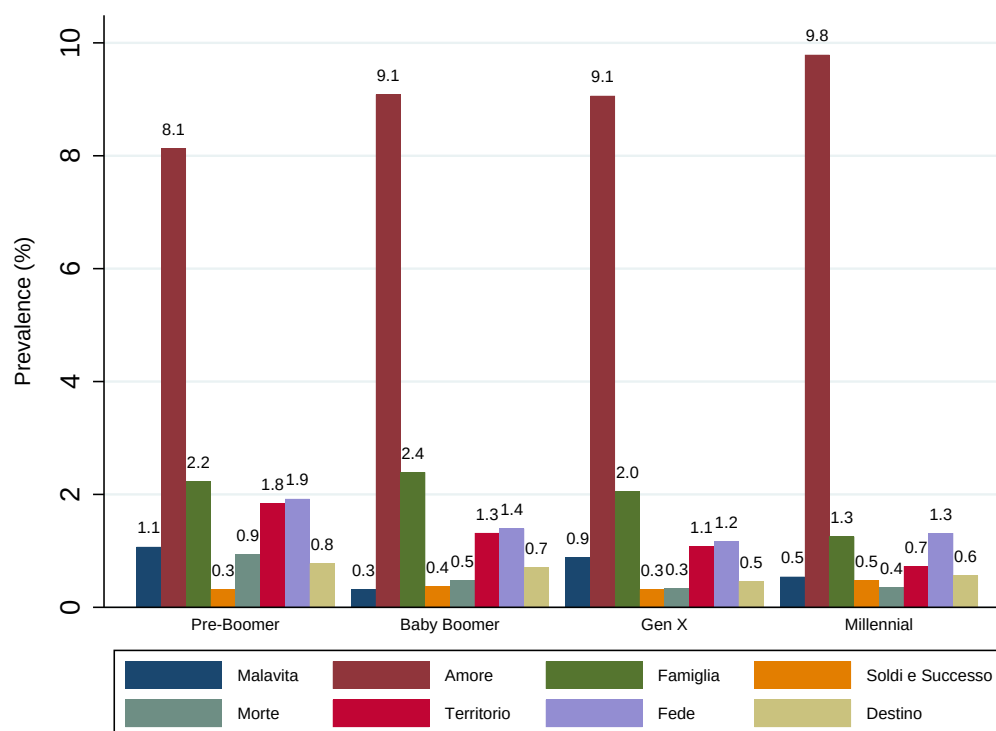
Interpretation: Figure 10 shows that Love remains the dominant title-based theme across all main cohorts. For the other themes, the graph suggests persistence and level differences, but it should not be pushed beyond its descriptive nature: classification is title-based and the dictionaries do not all have the same precision.

Table 2: Mean differences by generation for title-based macro-themes (Pre-Boomer, Baby Boomer, Gen X, Millennial)

Theme	PreB	Boom	GenX	Mill	1-2	1-3	1-4	2-3	2-4	3-4
Malavita	1.1	0.3	1.0	0.6	0.000	0.508	0.001	0.000	0.079	0.018
Love	8.1	8.9	9.0	9.9	0.076	0.029	0.000	0.713	0.056	0.110
Family	2.0	2.3	2.1	1.3	0.131	0.682	0.000	0.333	0.000	0.001
Money and Success	0.4	0.4	0.3	0.5	0.702	0.658	0.183	0.476	0.341	0.124
Death and Mourning	0.9	0.5	0.4	0.3	0.002	0.000	0.000	0.159	0.181	0.937
Territory	1.7	1.3	1.1	0.7	0.039	0.000	0.000	0.213	0.003	0.067
Faith	1.8	1.5	1.1	1.2	0.060	0.000	0.002	0.043	0.189	0.636
Destiny	0.9	0.7	0.5	0.4	0.278	0.000	0.001	0.037	0.045	0.900

Stata/Python code: `02_content_analysis_tracks.do`, final saving block on the canonical layer `tracks_unique_title_level.dta`; `export_track_report_tables.py` computes pairwise cohort comparisons and writes `track_theme_generation.table.tex`.

Figure 10: Prevalence of the 8 Macro-Themes by Generation



Notes: The Y-axis indicates theme incidence on titles for each main cohort. Love remains the dominant theme. **Stata code:** `02_content_analysis_tracks.do`, lines 127–139. `graph bar mt100... if inlist(gen_order,1,2,3,4), over(gen_order) ...`

7.4 Sentiment Analysis by Macro-Theme

Rationale: This analysis aims to see whether titles associated with different macro-themes shift over time toward a slightly more positive or more negative lexical balance.

Measures and Sentiment Estimation: For *tracks* as well, sentiment is now built with the same logic used in the lyrics report: tokenization of the text, counting of positive and negative lexical matches, and normalization by the number of observed tokens. In the case of *tracks*, the text unit is the song title:

$$\text{Sentiment Score} = \frac{\text{Positive Count} - \text{Negative Count}}{\text{Number of Tokens}} \quad (1)$$

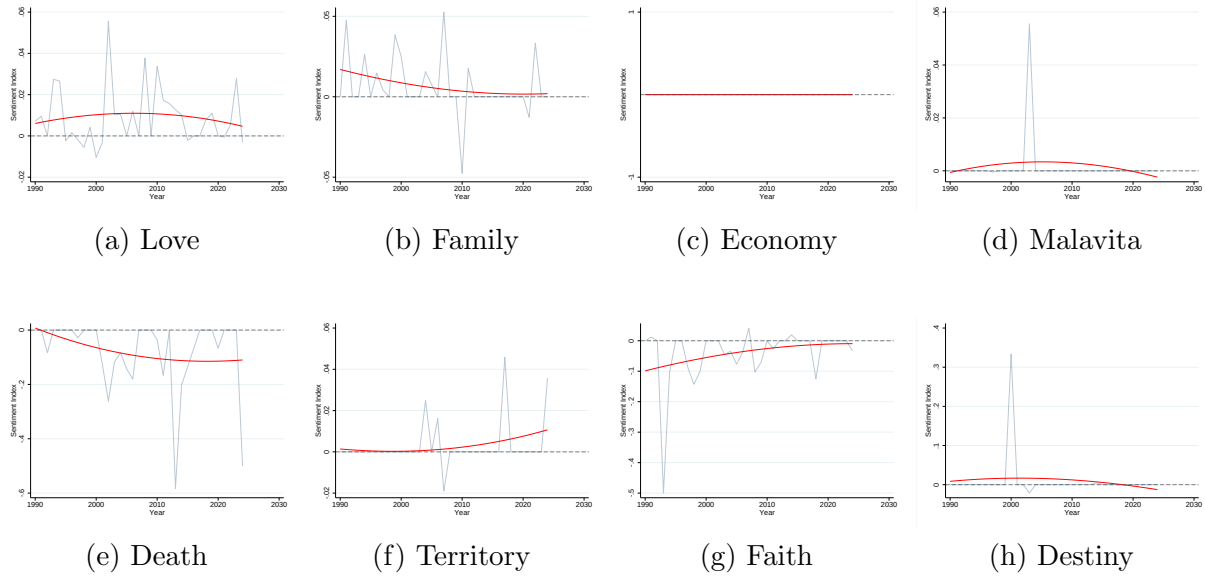
where the partial counts are defined on cleaned tokens through a conservative affective lexicon:

- **Positive Count:** matches to prefixes *felic, gioi, allegr, sorrid, seren, dolc, tener, sper, pace, abbracc, baci, carezz, desider, fiduc*.
- **Negative Count:** matches to prefixes *trist, dolor, soffer, piang, piant, lacrim, odi, rabb, paur, solitud, disper, trad, infam, ferit, maledett, colp, rimpiant*.

The final score remains a title-based lexical indicator, but it is now methodologically aligned with the lyrics one: not a simple binary presence/absence on the whole title, but a per-token balance.

Interpretation: Figure 11 should be read as a title-based heuristic of lexical polarity. The sentiment score is built with a very simple binary dictionary on titles; it therefore signals coarse lexical tone differences, not a refined measure of emotionality. Within this framework, the panels help compare relative patterns across themes, but they do not justify a strong psychometric reading.

Figure 11: Temporal Sentiment Trends for the 8 Macro-Themes (1990–2024)



Notes: The Y-axis reports the per-token sentiment balance in the title. The red line represents the local polynomial trend (lpoly).

Stata code: 02_content_analysis_tracks.do, lines 63–120 and 261–275. Snippet: `split/reshape of tokens; gen sentiment_score = (sent_pos_count - sent_neg_count)/sent_token_n; collapse (mean) sentiment_score, by(release_year)`

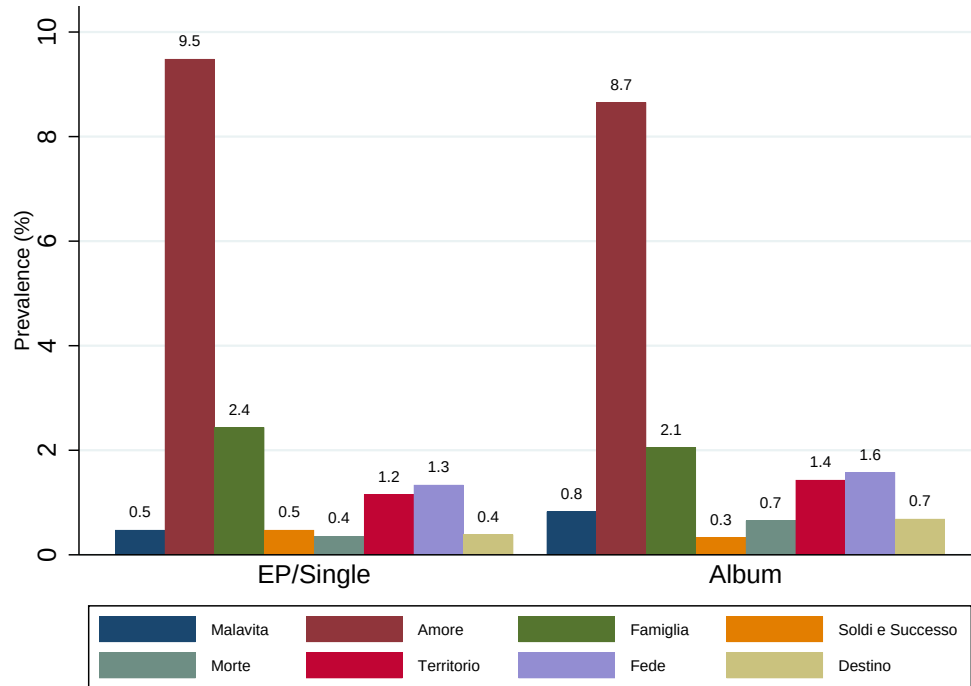
7.5 Thematic Prevalence: Album vs EP

Rationale: The choice of release format (traditional Album vs Single/EP) reflects different strategies of artistic communication and marketing. Analyzing whether certain themes vary in prevalence depending on format helps us understand whether EPs/singles function as vehicles for immediate-impact messages relative to the longer and more structured narrative of albums.

Measures: Percentage incidence of the 8 macro-themes stratified by release type (Album vs EPs/Singles).

Interpretation: Figure 12 suggests thematic differentiation between EPs/singles and albums, with a higher relative incidence of *Love* and *Family* in the former and a greater presence of *Malavita*, *Death*, and *Territory* in the latter. Here too the correct reading remains descriptive, since thematic classification is based on titles.

Figure 12: Comparison of Thematic Prevalence between Albums and EPs (8 Themes)



Notes: The Y-axis indicates the incidence of macro-themes on the total number of tracks by format. EPs show relative prevalence in Love and Family, whereas Albums focus more on Malavita, Death, and Territory.

Stata code: 02_content_analysis_tracks.do, lines 236-244. Snippet: `graph bar mt100..., over(rel_type) asyvars ...; graph export prev_format.pdf`

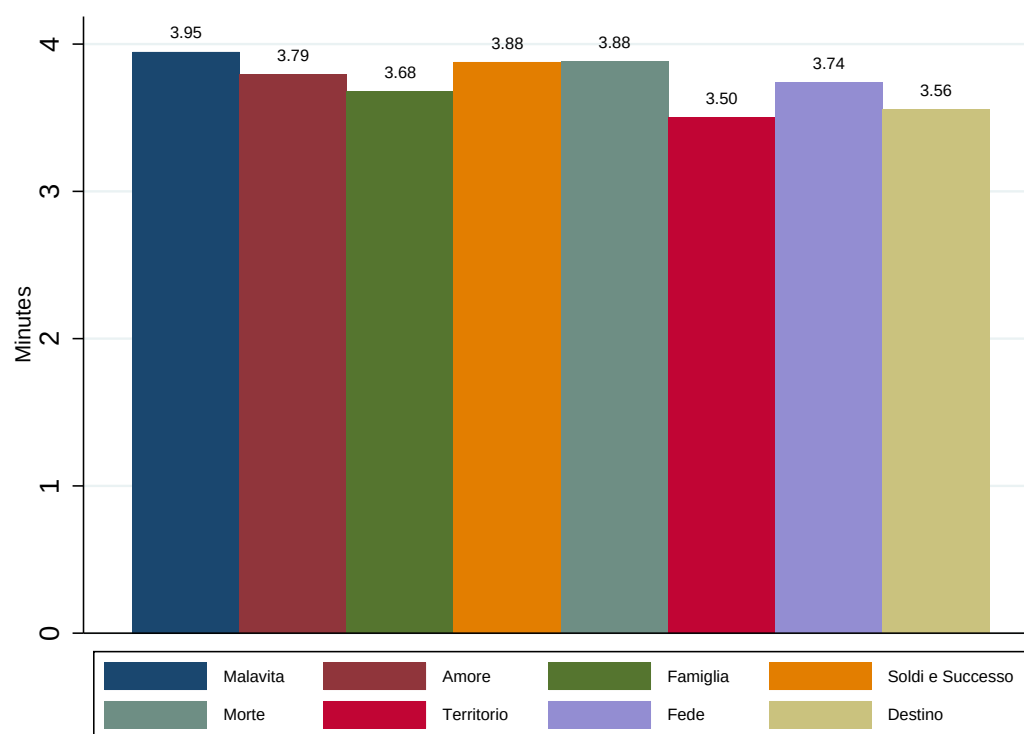
7.6 Average Duration by Main Theme

Rationale: Checking whether theme complexity affects track duration helps understand the underlying narrative structure (e.g., epic-narrative songs vs pop songs).

Measures: Mean duration in minutes by macro-theme, calculated exclusively on tracks released in Album format.

Interpretation: Figure 13 shows that average duration varies depending on the title-based macro-theme addressed in album tracks. This section describes differences in average running time, not differences in “narrative complexity” in any strong sense.

Figure 13: Average Song Duration (minutes) by the 8 Macro-Themes



Notes: The Y-axis reports average duration (minutes) of Album tracks. The Territory theme has the shortest average duration, while Death and Mourning records the highest average duration.

Stata code: 02.content.analysis.tracks.do, lines 246–259. keep if rel.type == 2; gen dur.malavita = duration_min if mt.malavita == 1; graph bar (mean) dur.... ...

A Appendix: Keywords and Descriptive Statistics

Table 3: Top 10 artists by track volume in the deduplicated corpus

Artist	Number of Tracks
mario merola	1,484
enzo di domenico	1,427
mauro nardi	1,057
mario trevi	925
gaetano setola	893
tony astarita	823
maria nazionale	714
tony bruni	704
gianluigi	658
franco moreno	600

Stata code: 01_exploratory_tracks.do, lines 55–82. `contract name, percent(pct)`
`freq(count); gsort -count; file write ... track_top_artists_table.tex`

Table 4: Absolute frequency and incidence of macro-themes in the title-based corpus

Macro-Theme	Count (N)	Incidence (%)
Malavita	266	0.82
Love	2,828	8.70
Family	648	1.99
Money and Success	121	0.37
Death and Mourning	199	0.61
Territory	429	1.32
Faith	484	1.49
Destiny	223	0.69
Thematic Coverage	4,974	15.30

Methodological note: Percentages do not sum to 100 because a track may activate multiple themes simultaneously and many tracks activate no dictionary at all.

Stata/Python code: 02_content_analysis_tracks.do, final saving block on the canonical layer `tracks_unique_title_level.dta`; `export_track_report_tables.py` computes counts, incidences, and thematic coverage and writes `track_topic_freq_table.tex`.

Table 5: Most frequent title words after lexical cleaning

Word	Freq.
amore	1,589
core	354
napule	331
bella	318
notte	298
vita	298
bene	274
mare	237
amo	218
mamma	216
canzone	213
vivo	203
luna	185
storia	185
sole	185

Stata/Python code: `02_content_analysis_tracks.do`, lines 177–233, exports the CSV with top words; `export_track_report_tables.py` reads `top_words_general.csv` and writes `track_top_words_table.tex`.

Table 6: Semantic Mapping of the 8 Macro-Themes

Macro-Theme	Sub-Topic	Main Keywords
Malavita	Omertà, Respect,	rispett, onor, omert, silenz, zitt, scapp, polizi, carabinier,
	Fugitivity, Police	divis, siren
Love	Love, Passion	ammore, amore, cor, cuor, innamor, vas, baci, passiu
Family	Blood ties	mamm, pate, padr, figl, frat, sor, nonn, nipot
Economy	Money, Success	sold, denar, ricch, cash, success, popolar, vinc, campion
Death	Mourning and Loss	mort, defunt, funer, chiagn, lacrim, bara, tomba, cimiter
Territory	Local identity	napule, napoli, vesuvio, mergellina, campania, barese, sicilia, palermo
Faith	Religiosity	gesu, madonna, sant, dio, preg, fede, carit, chies
Destiny	Life and Fortune	destin, sort, sciort, fatal, strada, cammin

Stata code: `02_content_analysis_tracks.do`, lines 11–57. `gen t_omerta = regexm(lower(title_clean), omerta_regex); gen mt_malavita = (t_omerta | t_rispetto | t_latitanza | t_polizia); gen mt_soldi = (t_soldi | t_successo) ...`